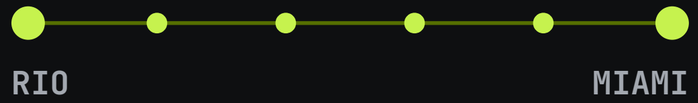
TESTNET · TEST KEYS · NOT REAL CUSTODIANS

Radox Custody Rail

Know who holds the goods: every handover signed by both custodians.

Toward a SWIFT-like custody standard on Stellar · Pilot corridor: Rio–Miami

Scan: the live shipment
rio-miami-rail.vercel.app



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PMF · THE PROBLEM

Who holds the cargo right now?

Warehouse — Port — Carrier — Customs — Recipient

45M

ocean bills of lading issued a year

DCSA, Feb 2023 (2021 data)

~11%

of them electronic in 2025

DCSA, figure as of 2025

22%

of global cargo thefts in 2025 involved insiders

TT Club & BSI Cargo Theft Report, Apr 2026

Context: our explainer video about a Rio–Miami trade-corridor initiative poses one possibility: a unique digital record for every shipment that both sides can check. This project is a custody layer for that record.

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PMF · WHAT IT COSTS

Cargo goes missing. Banks get fooled.

Our hypothesis: a financier cannot see who holds the goods before paying. No customer has confirmed it yet.

R$900M

est. direct loss, 8,570 cargo thefts in Brazil, 2025

NTC&Logística survey, 28 Mar 2026

US$85M

HSBC's loss on forged documents for oil trades that never happened

Hin Leong case, Global Trade Review, 18 Nov 2024

US$2.5T

global trade finance gap

ADB Global Trade Finance Gap Survey, Dec 2025

These numbers size the problem in the market, not results of ours. This build does not solve all of it.

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THE SOLUTION

Both custodians sign, or nothing changes

Outgoing custodian
`signs`

—

One Stellar transaction
`custody_registry.handover`

—

Incoming custodian
`signs`

Either signature missing → the call fails.

— Each handover moves custody and extends an on-chain hash chain, the `head`.

— The token stays with its holder in transit; only custody changes.

— After Delivered, the operator burns the token with the existing `redeem`[1].

A custody record keyed to the container number. In production it would sit beside the carrier's bill of lading; not integrated in this build.

[1] Radox platform policy, not enforced on-chain by the registry; redeem still needs the holder's signature.

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THE PRODUCT · LIVE ON TESTNET

7 testnet transactions. Anyone can check them.

Warehouse → Port, both signed	ledger 5381142 · 635c99…f0bd38
Port → Carrier, both signed	ledger 5381143 · 56a0f0…7b7cc2
Event attested by Source	ledger 5381144 · 333792…068905
Carrier → Customs, both signed	ledger 5381145 · ac268a…c3c4b2
Customs → Recipient, both signed	ledger 5381146 · 5199d4…400b7a
Delivered by Recipient to Holder, both signed	ledger 5381147 · 919def…144e37
Redeemed: token burned	864fd1…15644d

Verify head

`2b3d51f0` open · operator
`635c99ac` handover 1 · Warehouse → Port
`56a0f006` handover 2 · Port → Carrier
`ac268a49` handover 3 · Carrier → Customs
`5199d4fb` handover 4 · Customs → Recipient
`919def8c` deliver · Recipient → Holder
`864fd164` burn · existing redeem

7 successful

9 refused (simulated)

24 unit tests

MATCH head check, 25 Oct

`custody_registry`
`CCBJVXMJYWPHTDJHDFPRGKPGXGEP24JOPJAMUS5C3BMZJTRTWR2H4LII`

Plus attestation 33379211. Second run (branch three-machine): two team members authorised handovers on their own computers and testnet keys.

All test keys, mostly in one harness: it proves the mechanism, not independent custodians.

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ECOSYSTEM · WHY STELLAR

Why Stellar: two signers, one call

01	**Multi-party auth in one call**	Soroban authorises both custodians inside one contract invocation: the handover is atomic.
02	**Events plus an on-chain head**	RPC keeps contract events about 7 days. The head stored in the contract lets anyone rebuild the full history later.
03	**The token stays put, then burns**	The shipment token never moves in transit; the existing Radox redeem burns it after delivery.

Nearest neighbours we found on Stellar (LumenLoop, SCF, Scout, 24–25 Oct): Kutana and Trustless Work release escrow on trade milestones; Pipeline finances commodity trade. We found none that records a custody handover signed by two custodians, and no SEP.

Open source, Apache-2.0: a custody primitive other Stellar real-world-asset issuers can reuse.

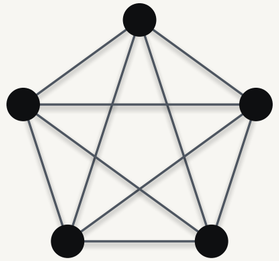
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USERS · BUSINESS MODEL

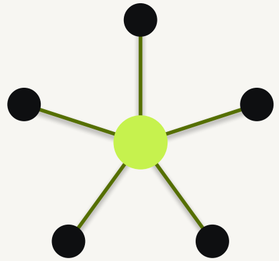
Integrate once, reach every party that joins

With one public custody standard, a port, airport, carrier, chamber or financier would integrate once and reach every other party: adopt without a custom project.

PAIR BY PAIR · 10 LINKS



ONE RADOX RAIL · 5 LINKS



Example, 5 parties: 10 custom links, or 5 links to one standard.

OPEN · ANYONE

— Record format, roles and events: the custody_registry contract (Apache-2.0)

— Verify any shipment, no login (slide 5)

RADOX OPERATES · CLIENT PAYS

— Custodian onboarding, issuance controls, operation

— The API and backend: closed, not self-hostable

— Paid by the financial institution; not priced

Adoption still takes work: a key per party, its own signatures, one connection to the rail (the Radox API).
Radox is the rail, never the issuer. Regulation and licences stay with the client and its local counsel.

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HACKATHON PROGRESS

Built this weekend, on disclosed prior work

PRIOR WORK · BEFORE SUN 25 OCT 09:00

— Radox platform (closed backend); its 3 Soroban contracts are on Stellar mainnet, public since 2026-08-19

— transfer_agent and its redeem (new testnet instance from the existing wasm)

— Architecture notes (20 Oct), planning files (24–25 Oct), the explainer video, our agent tooling (from 22 Oct)

— No contract code before then: first repo commit 25 Oct 09:00 UTC

BUILT AT HACKMERIDIAN

— custody_registry contract, 24 unit tests

— Lifecycle harness: 7 testnet transactions, 9 refused calls

— verify-head script, the public shipment page, this deck

— Radox API reads the registry: same head (closed backend, testnet)

AI attribution Code, tests, scripts, the shipment page, docs and this deck were made with AI coding agents (Claude Code); the team decided scope and what ships. Per-file record in AI_USAGE.md. The video narration is an AI voice generated from Pedro's voice.

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VISION · NEXT · THE ASK

Vision: one custody standard, every corridor

Rio–Miami is our pilot corridor. The goal is a public, SWIFT-like custody standard on Stellar for tokenized goods in transit worldwide, to integrate with the systems of chambers of commerce, ports and airports.

NOW

Mechanism proven on testnet. No standard, users or revenue yet.

NEXT

One Rio–Miami pilot shipment (none agreed yet). Full test matrix.

PLANNED

SCF Build, then a security audit (neither done). Mainnet.

FUTURE

More corridors; port, airport, chamber links (none confirmed); USD leg (not built).

Our ask **Introductions to one financial institution and one port, airport or logistics operator for a first pilot shipment, and a Stellar engineer's review of the design.**

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THE TEAM

Why us: a founder already on Stellar mainnet

Pedro Wakigawa Saragossy

Founder, captain. Built Radox: 3 Soroban contracts on Stellar mainnet.

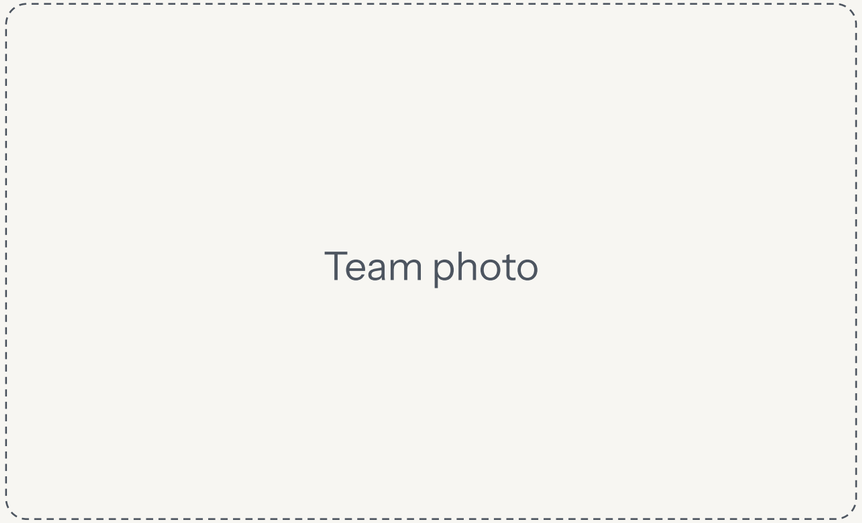
Leo Schlanger

Developer: custody_registry, the testnet harness and the proofs.

Vhendala

QA and product review: reads every deliverable as a judge would.

Leo and Vhendala joined for the hackathon.



Rio–Miami first. Our goal: one custody standard for every corridor.

Live page rio-miami-rail.vercel.app **Code** github.com/pwsaragossy/rio-miami-rail (Apache-2.0)

AI Code, page, docs and deck made with AI coding agents (Claude Code). Narration: an AI voice from Pedro's voice. No AI-generated images in this deck.

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